Analysis of Prompting Techniques in LLM Reasoning

GENAI EX-5.ipynb

AIM:

To Develop a Python program to analyze how different prompting strategies—Zero-shot, Few-shot, and Chain-of-Thought (CoT)—affect the reasoning capabilities of Large Language Models (LLMs). The program will also identify patterns in logical fallacies or hallucinations using benchmark datasets.

CODE:

```
# Install once
!pip install -q transformers datasets pandas matplotlib
google-generativeai sentencepiece

import re, time
import pandas as pd
import matplotlib.pyplot as plt
from datasets import load_dataset
from transformers import pipeline
import google.generativeai as genai
from getpass import getpass

# Gemini key
genai.configure(api_key=getpass("Enter Gemini API Key: "))

# Small dataset
data = load_dataset("gsm8k", "main")["test"].select(range(2))

# Models
hf = pipeline("text-generation", model="distilgpt2")
gm = genai.GenerativeModel("gemini-2.5-flash")

# Prompts
def zero(q): return f"Q: {q}\nA:"
def few(q): return f"Q: 2+3?\nA: 5\nQ: 10-4?\nA: 6\nQ: {q}\nA:"
def cot(q): return f"Solve step by step.\nQ: {q}\nA:"
```

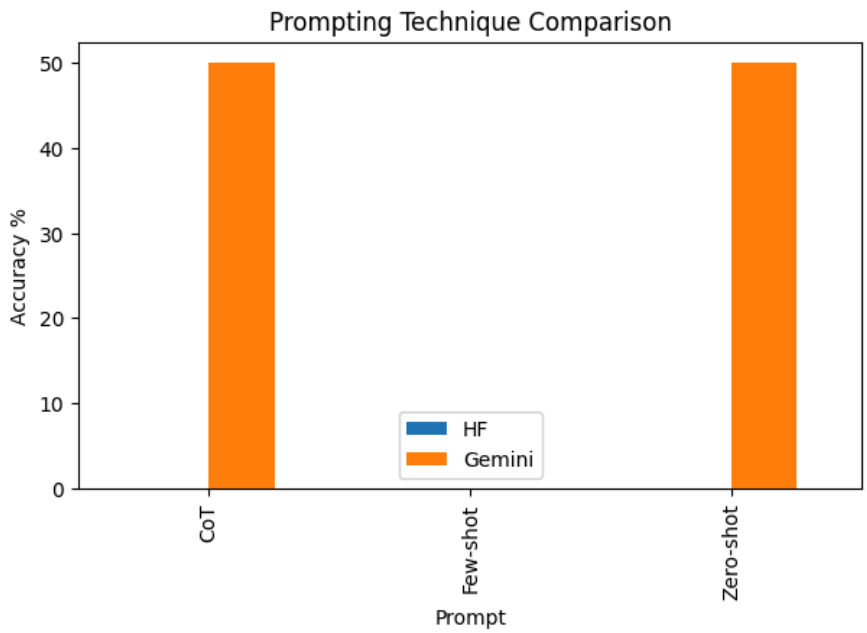
```python
# Query functions
def ask_hf(p):
    try:
        return hf(p, max_new_tokens=40, do_sample=False,
pad_token_id=50256)[0]["generated_text"]
    except:
        return "HF Error"

def ask_gm(p):
    try:
        return gm.generate_content(p).text
    except:
        return "Quota exceeded / Try later"

# Number extraction
def get_num(x):
    n = re.findall(r"-?\d+\.?\d*", str(x).replace(",", ""))
    return n[-1] if n else None

def get_gold(x):
    m = re.search(r"####\s*(-?\d+\.?\d*)", x.replace(",", ""))
    return m.group(1) if m else get_num(x)

# Result check
def check(pred, gold):
    if pred is None:
        return "Hallucination"
    elif pred == gold:
        return "Correct"
    else:
        return "Wrong"

# Run
res = []

for r in data:
    q = r["question"]
    gold = get_gold(r["answer"])

    for name, fn in [("Zero-shot", zero), ("Few-shot", few), ("CoT", cot)]:
```

```
        p = fn(q)

        hf_out = ask_hf(p)
        gm_out = ask_gm(p)
        time.sleep(10)

        hf_pred = get_num(hf_out)
        gm_pred = get_num(gm_out)

        res.append([
            name, q[:50], gold,
            hf_pred, gm_pred,
            check(hf_pred, gold),
            check(gm_pred, gold)
        ])

# Table
df = pd.DataFrame(res, columns=[
   "Prompt", "Question", "Gold", "HF_Pred", "Gemini_Pred", "HF_Result",
"Gemini_Result"
])

print(df)

# Accuracy
acc = pd.DataFrame({
   "HF": df["HF_Result"].eq("Correct").groupby(df["Prompt"]).mean() * 100,
   "Gemini":
df["Gemini_Result"].eq("Correct").groupby(df["Prompt"]).mean() * 100
})

print("\nAccuracy (%)")
print(acc)

# Plot
acc.plot(kind="bar", figsize=(7,4))
plt.title("Prompting Technique Comparison")
plt.ylabel("Accuracy %")
plt.show()
```

```python
# Summary
print("\nHF Summary")
print(df["HF_Result"].value_counts())

print("\nGemini Summary")
print(df["Gemini_Result"].value_counts())
```

OUTPUT:

```
       Prompt                                          Question Gold HF_Pred
0  Zero-shot  Janet’s ducks lay 16 eggs per day. She eats th...   18       2
1   Few-shot  Janet’s ducks lay 16 eggs per day. She eats th...   18       5
2        CoT  Janet’s ducks lay 16 eggs per day. She eats th...   18       2
3  Zero-shot  A robe takes 2 bolts of blue fiber and half th...    3     1.5
4   Few-shot  A robe takes 2 bolts of blue fiber and half th...    3      -4
5        CoT  A robe takes 2 bolts of blue fiber and half th...    3     1.5

  Gemini_Pred HF_Result  Gemini_Result
0        None     Wrong  Hallucination
1        None     Wrong  Hallucination
2        None     Wrong  Hallucination
3           3     Wrong        Correct
4        None     Wrong  Hallucination
5           3     Wrong        Correct

Accuracy (%)
            HF  Gemini
Prompt
CoT        0.0    50.0
Few-shot   0.0     0.0
Zero-shot  0.0    50.0
```



HF Summary
HF_Result
Wrong 6
Name: count, dtype: int64

Gemini Summary
Gemini_Result
Hallucination 4
Correct 2
Name: count, dtype: int64